Numerical Problems based on the hardness of water:

Problem-1: A sample of water is found to contains following dissolving salts in milligrams per litre $Mg(HCO_3)_2$ = 73, $CaCl_2$ = 111, $Ca(HCO_3)_2$ = 81, $MgSO_4$ = 40 and $MgCl_2$ = 95. Calculate temporary and permanent hardness and total hardness.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (mg/Lit)
$Mg(HCO_3)_2$	73	146	73×100/146 = 50
$CaCl_2$	111	111	111×100/111 = 100
$Ca(HCO_3)_2$	81	162	81×100/162 = 50
$MgSO_4$	40	120	40×100/120 = 33.3
$MgCl_2$	95	95	95×100/95 = 100

Temporary hardness = $Mg(HCO_3)_2$ + $Ca(HCO_3)_2$

= 50 + 50 = 100mgs/Lit.

Permanent hardness = $CaCl_2$ + $MgSO_4$ + $MgCl_2$

= 100 + 33.3 + 100 = 233.3mgs/Lit.

Total hardness = Temporary hardness + Permanent hardness

= 100 + 233.3 = 333.3mgs/Lit.

Problem-2: A sample of water is found to contains following dissolving salts in milligrams per litre $Mg(HCO_3)_2$ = 16.8, $MgCl_2$ = 12.0, $MgSO_4$ = 29.6 and NaCl = 5.0. Calculate temporary and permanent hardness of water.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (mg/Lit)
$Mg(HCO_3)_2$	16.8	146	16.8×100/146 = 11.50
$MgCl_2$	12.0	95	12.0×100/95 = 12.63
$MgSO_4$	29.6	120	29.6×100/120 = 24.66
NaCl	5.0	NaCl does not contribute any hardness to water hence it is ignored.	

Temporary hardness = $Mg(HCO_3)_2$ = 11.50mgs/Lit.

Permanent hardness = $MgCl_2$ + $MgSO_4$ = 12.63 + 24.66 = 37.29mgs/Lit.

Problem-3: A sample of water is found to contains following analytical data in milligrams per litre $Mg(HCO_3)_2$ = 14.6, $MgCl_2$ = 9.5, $MgSO_4$ = 6.0 and $Ca(HCO_3)_2$ = 16.2. Calculate temporary and permanent hardness of water in parts per million, Degree Clarke's and Degree French.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(mg/Lit)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (mg/Lit)
$Mg(HCO_3)_2$	14.6	146	14.6×100/146 = 10
$MgCl_2$	9.5	95	9.5×100/95 = 10
$MgSO_4$	6.0	120	6.0×100/120 = 5
$Ca(HCO_3)_2$	16.2	162	16.2×100/162 =10

Temporary hardness [Mg $(HCO_3)_2$ + Ca $(HCO_3)_2$] = 10 + 10 = 20mg/Lit
= 20ppm
= 20×0.07°Cl = 1.4°Cl
= 20×0.1°Fr = 2°Fr

Permanent hardness [$MgCl_2$ + $MgSO_4$] = 10 + 5 = 15mg/Lit
= 15ppm
= 15×0.07°Cl = 1.05°Cl
= 15×0.1°Fr = 1.5°Fr

Problem-4: Calculate the amount of temporary and permanent hardness of a water sample in Degree Clarke's, Degree French and Milligrams per Litre which contains following impurities.
$Ca(HCO_3)_2$ = 121.5 ppm, $Mg(HCO_3)_2$= 116.8 ppm, $MgCl_2$ = 79.6 ppm and $CaSO_4$ = 102 ppm.

Solution:

Name of the hardness causing salts	Amount of the hardness causing salts(ppm)	Molecular weight of hardness causing salts	Amounts equivalent to $CaCO_3$ (ppm)
$Ca(HCO_3)_2$	121.5	162	121.5×100/162 = 75
$Mg(HCO_3)_2$	116.8	146	116.8×100/146 = 80
$MgCl_2$	79.6	95	79.6×100/95 = 3.37
$CaSO_4$	102	136	102×100/136 = 75

Temporary hardness [Mg $(HCO_3)_2$ + Ca $(HCO_3)_2$] = 75 + 80 = 155 ppm
= 155×0.07°Cl = 10.85°Cl
= 155×0.1°Fr = 15.5°Fr
= 155×1mg/Lit = 155 mg/Lit

Permanent hardness [$MgCl_2$ + $CaSO_4$] = 10 + 5 = 15mg/Lit
= 15ppm
= 15×0.07°Cl = 1.05°Cl
= 15×0.1°Fr = 1.5°Fr

Problem-5: Calculate the temporary and permanent hardness of water sample containing $Mg(HCO_3)_2$= 7.3mg/L, $Ca(HCO_3)_2$= 16.2mg/L, $MgCl_2$= 9.5mg/L, $CaSO_4$=13.6mg/L).

Solution: conversion into $CaCO_3$ equivalents:

Constituent	Multiplication factor	$CaCO_3$ equivalent
$Mg(HCO_3)_2$= 7.3mg/L	100/146	7.3X100/146= 5mg/L
$Ca(HCO_3)_2$= 16.2mg/L	100/162	16.2X100/162=10mg/L
$MgCl_2$= 9.5mg/L	100/95	9.5X100/95= 10mg/L
$CaSO_4$=13.6mg/L	100/136	13.6X100/136= 10mg/L

∴ Temporary hardness of water due to $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$ =
=5+10=15mg/L or 15ppm.

Permanent hardness due to $MgCl_2$ and $CaSO_4$ = 10+10=20mg/L or 20ppm.

Problem-6: **Calculate the temporary and total hardness of a water sample containing $Mg(HCO_3)_2$= 73mg/L, $Ca(HCO_3)_2$= 162mg/L, $MgCl_2$= 95mg/L, $CaSO_4$=136mg/L.**

Solution: calculation of $CaCO_3$ equivalents:

Constituent	Multiplication factor	$CaCO_3$ equivalent
$Mg(HCO_3)_2$= 73mg/L	100/146	73X100/146= 50mg/L
$Ca(HCO_3)_2$= 162mg/L	100/162	162X100/162=100mg/L
$MgCl_2$= 95mg/L	100/95	95X100/95= 100mg/L
$CaSO_4$=136mg/L	100/136	136X100/136= 100mg/L

$\therefore$ Temporary hardness of water due to $Mg(HCO_3)_2$ and $Ca(HCO_3)_2$ =

=100 + 50=150mg/L or ppm.

Total hardness of water= 50+100+100+100=350 mg/L or ppm.

Exercises for Practice:

1. **A sample of water contains 18 mg/L of $Ca(HCO_3)_2$, 17 mg/L of $CaSO_4$, 12 mg/L of $CaCl_2$, 14 mg/L of $Mg(HCO_3)_2$, and 14 mg/L of $MgCl_2$. Calculate temporary, permanent and total hardness in terms of $CaCO_3$ equivalent. Given molecular weight $Ca(HCO_3)_2$= 162, $CaSO_4$= 136, $Mg(HCO_3)_2$= 146, $MgCl_2$= 95.**
2. **A sample of water contains 22 mg/L of $Ca(HCO_3)_2$, 17 mg/L of $CaSO_4$, 11 mg/L of $CaCl_2$, 32 mg/L of $Mg(HCO_3)_2$, and 21 mg/L of $MgCl_2$. Calculate temporary, permanent and total hardness in terms of $CaCO_3$ equivalent. Given molecular weight $Ca(HCO_3)_2$= 162, $CaSO_4$= 136, $Mg(HCO_3)_2$= 146, $MgCl_2$= 95.**
3. **A sample of water contains 20 mg/L of $Ca(HCO_3)_2$, 16 mg/L of $CaSO_4$, 12 mg/L of $CaCl_2$, 11 mg/L of $Mg(HCO_3)_2$, and 21 mg/L of $MgCl_2$. Calculate temporary, permanent and total hardness in terms of $CaCO_3$ equivalent. Given molecular weight $Ca(HCO_3)_2$= 162, $CaSO_4$= 136, $Mg(HCO_3)_2$= 146, $MgCl_2$= 95.**
4. **A sample of water contains 20 mg/L of $Ca(HCO_3)_2$,30 mg/L of $CaSO_4$, 12 mg/L of $CaCl_2$, 14 mg/L of $Mg(HCO_3)_2$, and 56 mg/L of $MgCl_2$. Calculate temporary, permanent and total hardness in terms of $CaCO_3$ equivalent.**

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